Dataset of FDM 3D Printing Process Parameters and Mechanical Properties for Multi-Material Characterization

Description

This dataset contains comprehensive experimental data on the fused deposition modeling (FDM) 3D printing process, focusing on the relationship between manufacturing parameters and the resulting physical and mechanical properties of printed parts. It comprises 500 unique experimental runs/samples and covers a diverse range of polymer materials and printing configurations.

The dataset is highly valuable for researchers in additive manufacturing, materials science, and machine learning practitioners looking to predict printed part quality, optimize manufacturing processes, or build surrogate models for mechanical behavior.

Dataset Structure & Parameters:

1. Input / Process Parameters:

- **Printer Type:** Open vs. Closed chamber printers.
- **Material Types (10 different filaments):** PLA+, PLA, PLA phosphor (fosfor), PLA-CF (carbon fiber), PLA-transparent (şeffaf), PETG, rPET (recycled PET), PP, TPU, and ABS.
- **Layer Thickness (mm):** Variable layer resolutions.
- **Infill Density (%):** Variable infill percentages.
- **Infill Pattern (15 different patterns):** Grid, Triangles, Zigzag, Lines, Cubic, Quarter Cubic, Cross, Gyroid, Cubic Subdivision, Tri-hexagon, Lightning, Cross3D, Concentric, and Octet.
- **Printing Speed (mm/s):** Controlled extrusion speeds.
- **Specimen Dimensions:** Width (mm) and Thickness (mm) of each printed sample.

2. Output / Characterization Results:

- **Surface Roughness (Roughness AVG - µm):** Average surface roughness measurements.
- **Hardness (Hardness AVG):** Shore hardness characterization.
- **Tensile Properties:** Peak Load (N), Peak Stress (kPa), Strain at Break (mm/mm), and Elastic Modulus (MPa).

Potential Applications:

- Optimizing 3D printing parameters for specific mechanical and surface quality performance targets.
- Comparative analysis of different filament types (e.g., standard PLA vs. carbon fiber reinforced PLA-CF or recycled rPET).
- Training and developing machine learning, deep learning, and regression models (such as ensemble learning models) to predict mechanical strength, surface roughness, or elasticity.

Associated Publication: If you use this dataset in your research, please cite our corresponding paper:

Aktepe, E., & Ergün, U. (2026). HMQ-ES-Stack-GBR: A Hybrid Ensemble Learning Model for Mechanical and Physical Quality Prediction in FDM 3D Printing. Micromachines.

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